

Question Bank (Reframed)

UNIT – I

1. Define Deep Learning and explain how it differs from Machine Learning with the help of a performance graph.
2. What is a perceptron? Describe its types with suitable examples.
3. Describe various loss functions used in regression and classification problems. 4. Illustrate any three activation functions commonly used in Deep Learning and explain their significance.
5. Discuss the following optimizers with their advantages and disadvantages: -
 - Stochastic Gradient Descent
 - Adaptive Gradient Descent
6. Compare Vanilla Gradient Descent and Mini-batch Gradient Descent, highlighting their benefits and drawbacks.
7. Explain the concept of overfitting in Deep Learning. What are its causes and strategies to prevent it?
8. Describe early stopping with the help of a graph and discuss its benefits.
9. List and describe the different types of generative models.
10. What is linear separability? Why can't the perceptron handle non-linear data? Justify with an example.
11. Write short notes on the following:
 - Variational Autoencoders (VAEs)
 - Generative Adversarial Networks (GANs)

UNIT – II

12. What is sequence data? List its key characteristics and provide examples. 13. Explain the problems faced by Artificial Neural Networks (ANN) when dealing with sequential data.
14. Describe the architecture and working of a Recurrent Neural Network (RNN) with a neat diagram.
15. Summarize the applications of Recurrent Neural Networks (RNN).
16. Explain the limitations and challenges faced by RNNs.
17. Compare the types of RNNs along with their applications.
18. Write a short note on Long Short-Term Memory (LSTM).
19. Explain the architecture of LSTM with a neat labeled diagram.
20. Describe the following components of LSTM:
 - States
 - Input Gate
 - Output Gate
21. Explain Seq2Seq data and Seq2Seq models with suitable examples.
22. Briefly describe the Encoder–Decoder architecture with an example.

23. Discuss the limitations of the Encoder–Decoder model and explain the Attention Mechanism.
24. Write a short note on the Transformer model.
25. What is self-attention and why is it important?

UNIT – III

26. Explain the visual cortex system of humans with a neat labeled diagram.
27. Write a short note on Convolutional Neural Networks (CNNs).
28. Explain the following components of CNN:
 - Kernel/Filter
 - Strides
29. Explain the following operations in CNN:
 - Sliding the filter
 - Padding
30. Explain the following operations in CNN:
 - Convolution operation
 - Pooling
31. Describe pooling in CNN and explain its types.
32. Justify why we cannot use ANN instead of CNN for image-related tasks.
33. Differentiate between Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN).
34. Explain the LeNet-5 architecture.
35. Write a short note on the VGG-16 model.
36. Explain the architecture of VGG-16 with a neat labeled diagram.
37. Describe the Residual Network (ResNet) and explain the challenges it resolves.
38. Briefly explain the concept of Transfer Learning.

UNIT – IV

39. Write a short note on Autoencoders.
40. Explain the architecture of an Autoencoder.
41. Briefly explain Denoising Autoencoders along with their advantages and disadvantages.
42. Briefly explain Variational Autoencoders along with their advantages and disadvantages.
43. List and explain different types of Autoencoder models.
44. Write a short note on Generative Adversarial Networks (GANs).
45. Explain the following components of GANs:
 - Generator
 - Discriminator
46. Describe the following challenges in training GANs:
 - Oscillation Loss
 - Mode Collapse
47. Write a short note on Deep Convolutional GANs (DCGANs).
48. Describe how DCGANs differ from vanilla GANs.
49. Explain the architecture of DCGANs.

50. List and explain any six hyperparameters of GANs.